

## SUMMARY TABLE: PHYSICS GRADE 10

*Sub-Strand 1.1: Pressure — Teacher Reference (Pre-filled)*

| SUMMARY TABLE: PHYSICS GRADE 10 |                                                                                                                                                                |
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| Sub-Strand                      | Sub-Strand 1.1: Pressure                                                                                                                                       |
| Driving Question                | DRIVING QUESTION: "Why did the jerrycan crush itself — and how does the invisible push of fluids move mountains, save lives, and power machines across Kenya?" |

| INSTRUCTIONS                                                                                                                                                                                                                                                                            |
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| FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson. |

| Lesson # and Title                                                                                                | What did I observe?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | What did I learn?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | How does this explain the phenomenon?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <b>Lesson 1</b><br><b>The Mystery Begins: Launching the Crushed Jerrycan Phenomenon</b>                           | Students observed a sealed jerrycan (or plastic bottle) being crushed inward after hot water was swirled inside and the cap was tightly closed. The visible, dramatic collapse of the container with no external force applied served as the anchoring phenomenon. Teacher note: Ensure all students can see the deformation clearly — consider repeating the demo or circulating the crushed can. Cold-call at least 4 students for initial predictions before revealing any explanation.                       | The atmosphere exerts a real, measurable force — called atmospheric pressure — that pushes inward on all surfaces from every direction. When the air pressure inside the jerrycan dropped (as steam condensed and cooled), the unbalanced external atmospheric pressure (~101.3 kPa) crushed the can inward. Teacher note: Emphasise that this is not 'suction' pulling the can in — it is the outside air PUSHING in. This misconception correction is critical and will resurface in Lessons 4 and 6.                                                                                | Students began constructing an initial model linking air pressure, force, and area. They recorded on the Driving Question Board (DQB) their best current explanation for the jerrycan crush and flagged questions they cannot yet answer (e.g., 'Why doesn't the air crush us too?', 'Does water also push like this?'). Teacher note: Do NOT resolve DQB questions yet — preserving productive uncertainty is pedagogically essential here. The crushed jerrycan should be displayed in the classroom throughout the unit as a visual anchor. |
| <b>Lesson 2</b><br><b>What Is Pressure? Measuring the Invisible Push — Shoes, Scales, and Force Per Unit Area</b> | Students compared the impressions made in soft ground (or a foam pad) by a flat shoe vs. a pointed heel carrying the same weight, and by an elephant's foot vs. a nail. They also investigated how a sharp panga cuts more easily than a blunt one despite the same applied force. Kenya context: Reference how Maasai warriors' sandals distribute weight across sand differently from pointed heels at a Nairobi office. Teacher note: If foam pads are unavailable, use diagrams or a simple pressure-on-sand | Pressure is defined as force per unit area: $P = F/A$ , measured in Pascals ( $\text{Pa} = \text{N/m}^2$ ). The same force produces HIGHER pressure when applied over a smaller area, and LOWER pressure when spread over a larger area. Students practised calculating pressure using sample values (e.g., a 60 kg student standing on one foot on a shoe sole of area $0.02 \text{ m}^2$ ). Teacher note: Verify students correctly convert units ( $\text{kg} \rightarrow \text{N}$ using $W = mg$ , $g = 10 \text{ N/kg}$ ). A common error is using mass instead of weight in the | Students revisited the jerrycan phenomenon: the atmosphere exerts ~101,300 N on every square metre of surface. They calculated the total force on one face of the jerrycan, making the 'invisible push' quantitatively real. DQB update: Students added new evidence to their board — the jerrycan crushed because the FORCE PER UNIT AREA outside was greater than inside. Teacher note: Prompt students to revise their initial models to include the $P = F/A$ relationship and                                                             |

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|                                                                                                                         | demonstration outdoors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | formula — address explicitly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | directional arrows showing inward atmospheric force.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Lesson 3</b><br><b>Pressure in Gases:</b><br><b>Particle Model and Boyle's Law</b>                                   | Students observed a syringe with the nozzle blocked: pushing the plunger in compresses the trapped air, and releasing it causes the plunger to spring back. They also observed a balloon being taken to high altitude (via video or diagram) expanding as external pressure drops. Kenya context: Reference how sealed food packaging puffs up when transported from Nairobi (1,680 m) to high-altitude Kericho or the Aberdares. Teacher note: Allow 3–4 student volunteers to feel the resistance when compressing the syringe — tactile experience reinforces the particle model.                                          | Gas pressure arises from collisions of gas particles with container walls. Compressing a gas into a smaller volume increases the frequency of particle collisions, raising pressure. At constant temperature, pressure and volume are inversely related (Boyle's Law: $P_1V_1 = P_2V_2$ ). Students practised Boyle's Law calculations. Teacher note: Ensure students connect the PARTICLE MODEL to the macroscopic behaviour — this underpins the explanation of the jerrycan collapse (fewer particles inside → fewer wall collisions → lower pressure).                                           | Students updated their jerrycan model: as steam condensed, the number of gas particles inside decreased, reducing particle-wall collisions and therefore internal pressure. The external atmospheric pressure remained ~101.3 kPa, creating a net inward force that crushed the can. DQB update: Students can now answer 'Why did the inside pressure drop?' — add this resolution to the board. Teacher note: Highlight that Boyle's Law applies only at constant temperature; the cooling of the jerrycan is a temperature change, making this a more complex real-world scenario — acknowledge this honestly. |
| <b>Lesson 4</b><br><b>The Invisible Giant:</b><br><b>Proving the Atmosphere Has Weight and Pushes in All Directions</b> | Students observed the classic 'water-in-inverted-glass' demonstration (a card holds water in an upside-down glass) and a rubber suction cup sticking to a smooth surface. They also examined a diagram of a mercury barometer and discussed how weather stations in Nairobi and Mombasa measure atmospheric pressure. Teacher note: Perform the inverted glass demo TWICE — once asking students to predict, then revealing. Cold-call at least 5 students for predictions. Ensure the Mombasa vs. Nairobi pressure comparison is discussed (Mombasa, at sea level, has higher atmospheric pressure than Nairobi at 1,680 m). | The atmosphere is a fluid with weight; this weight creates a pressure of approximately 101.3 kPa acting in ALL directions simultaneously — upward, downward, and sideways. Atmospheric pressure decreases with altitude because there is less air above. Students used the relationship $P = F/A$ to calculate the force the atmosphere exerts on everyday objects (a human body, a school desk, a jerrycan). Teacher note: Stress the 'all directions' concept — students commonly assume pressure only acts downward. The inverted glass demo is the key evidence for UPWARD atmospheric pressure. | Students revisited and refined their jerrycan model to explicitly show atmospheric pressure arrows acting inward from ALL sides — not just from the top. DQB update: The question 'Why doesn't the air crush us too?' can now be partially answered — our bodies have internal pressure (blood, fluids, and air in lungs) that balances the external atmospheric pressure. Teacher note: This is a major conceptual milestone. Celebrate it explicitly. Prompt students to consider: 'What would happen to your body at very high altitude or deep underwater?' (previewing Lesson 6).                           |
| <b>Lesson 5</b><br><b>Transmission of Pressure in Fluids — Pascal's Principle</b>                                       | Students observed a sealed bag of water being squeezed — water squirts equally from all holes punched around it. They also watched a video or diagram of a hydraulic car jack being used at a Nairobi garage to lift a matatu. A hands-on activity used two syringes of different sizes connected by a tube filled with water — pushing on the small syringe moved the large syringe's plunger with greater force. Teacher note: The two-syringe activity is high-impact — ensure every group completes it. Ask:                                                                                                              | Pressure applied to an enclosed (confined) fluid is transmitted equally and undiminished in ALL directions throughout the fluid — this is Pascal's Principle. In a hydraulic system, $F_1/A_1 = F_2/A_2$ , which allows a small input force on a small piston to generate a large output force on a large piston. Students derived and practised the hydraulic force formula. Teacher note: Emphasise the trade-off — a hydraulic system multiplies FORCE but not energy. The large piston moves a smaller distance                                                                                  | Students connected Pascal's Principle back to the jerrycan: when the internal pressure changed, it changed uniformly throughout the enclosed gas inside the can, causing uniform inward deformation. DQB update: Students added 'Pascal's Principle' as a new tool on the board — now they can explain hydraulic brakes in matatus and buses. Teacher note: Prompt model revision to include fluid transmission arrows. Ask students: 'How does a bus driver in Mombasa stop a 10-tonne vehicle                                                                                                                  |

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|                                                                                                               | 'Where have you seen a small force create a big force in Kenya?' (jikos, brakes, dental chairs).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | than the small piston. This energy conservation point prevents a common misconception about 'getting something for nothing.'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | with gentle foot pressure?' — this motivates the next lesson's deeper hydraulic analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Lesson 6</b><br><b>Liquid Pressure: Depth, Density, and the Formula <math>P = \rho gh</math></b>           | Students observed water squirting faster and farther from a hole near the BOTTOM of a water-filled container than from a hole near the top. They also examined diagrams showing how water pressure increases with depth in Lake Victoria, and how fish and divers experience increasing pressure as they descend. Kenya context: Reference how water towers in Nairobi are built tall to create sufficient pressure for household taps, and how the Gitaru Hydroelectric Power Station on the Tana River uses the pressure of deep water. Teacher note: The multi-hole water container demo is highly visual — allow students to observe and sketch the water jet trajectories before explaining why.  | Liquid pressure increases with depth ( $h$ ) and with the density ( $\rho$ ) of the liquid, and is calculated using $P = \rho gh$ , where $g = 10 \text{ N/kg}$ . The pressure at a given depth is the same in all directions and does not depend on the shape of the container. Students calculated pressure at various depths in water ( $\rho = 1000 \text{ kg/m}^3$ ) and compared it to pressure in a denser liquid like salt water ( $\rho \approx 1025 \text{ kg/m}^3$ , relevant to the Indian Ocean at Mombasa). Teacher note: Contrast $P = \rho gh$ (liquid pressure due to depth) with $P = F/A$ (pressure due to applied force) — students must know when to apply each formula. A worked example comparing a shallow Nairobi water tank vs. a deep Tana River dam is effective. | Students extended their model: liquid pressure (unlike gas pressure) is directional with depth, but still acts in all directions at any given point. They connected this to the jerrycan phenomenon by considering: if the jerrycan were filled with water and sealed, would depth pressure inside affect the outcome? DQB update: Students can now answer 'Does water push like air?' — YES, but liquid pressure increases with depth in a way gas pressure does not (under normal conditions). Teacher note: Ensure students update their models to distinguish between gas pressure (uniform throughout) and liquid pressure (increases with depth). |
| <b>Lesson 7</b><br><b>Pascal's Principle &amp; Hydraulic Systems — Pressure Transmitted, Power Multiplied</b> | Students examined real or pictured hydraulic systems: a matatu's hydraulic brake system, a hydraulic press at a Nairobi jua kali workshop, a dentist's adjustable chair, and agricultural hydraulic lifts used on large farms in the Rift Valley. They worked through a detailed two-syringe investigation, measuring force input and output and comparing predicted vs. actual values using $F_1/A_1 = F_2/A_2$ . Teacher note: If real hydraulic equipment is unavailable, a high-quality video of a hydraulic crane or JCB excavator operating at a Nairobi construction site is highly effective. Cold-call students to identify the 'small piston' and 'large piston' in each real-world example. | Pascal's Principle ( $P$ applied = $P$ transmitted) enables hydraulic machines to multiply force: $F_2 = F_1 \times (A_2/A_1)$ . The mechanical advantage equals the ratio of piston areas. Energy is conserved — the gain in force comes at the cost of a smaller displacement on the output side. Students solved multi-step problems involving hydraulic systems and identified hydraulic machines in Kenyan industry and medicine. Teacher note: This lesson deepens Lesson 5's introduction. Students should now solve quantitative problems independently. A common error is confusing the ratio — remind students that the LARGER piston produces the LARGER force, which corresponds to the larger area.                                                                              | Students fully articulated how a bus driver in Mombasa can stop a 15-tonne vehicle with modest foot pressure: the brake fluid transmits pressure equally to all four large wheel cylinders (Pascal's Principle), generating enormous braking force. DQB update: The question 'How does the invisible push of fluids power machines?' is now substantially answered — add hydraulic systems as a resolved entry. Teacher note: Prompt students to revise their cumulative models to include a hydraulic system diagram alongside their jerrycan and atmospheric pressure models, showing the unifying theme: FLUIDS TRANSMIT AND MULTIPLY PRESSURE.      |
| <b>Lesson 8</b><br><b>Pressure in Liquids: Deriving and Applying <math>P = \rho gh</math></b>                 | Students derived $P = \rho gh$ from first principles by considering the weight of a column of liquid above a point. They verified the formula experimentally using a transparent tube filled with liquids of different densities (water, salt water, coloured cooking oil) and measuring the                                                                                                                                                                                                                                                                                                                                                                                                           | Pressure in a liquid depends ONLY on: (1) the density of the liquid ( $\rho$ ), (2) gravitational field strength ( $g = 10 \text{ N/kg}$ ), and (3) the depth below the surface ( $h$ ). It does NOT depend on the volume of liquid or the shape of the container (the hydrostatic paradox). Students applied $P = \rho gh$ to calculate                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Students connected liquid pressure to the broader unit: the jerrycan collapsed because of ATMOSPHERIC pressure (a gas fluid), while dam walls are built thick at the base because of LIQUID pressure ( $P = \rho gh$ ). Both phenomena arise from the same fundamental idea — fluids exert pressure in                                                                                                                                                                                                                                                                                                                                                  |

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|                                                                                                                         | <p>pressure at equal depths — confirming that denser liquids exert greater pressure. Kenya context: Students calculated the water pressure at the base of the Kindaruma Dam on the Tana River and at the bottom of a standard Nairobi household water storage tank (1.5 m tall). Teacher note: The derivation (<math>P = F/A = mg/A = \rho Vg/A = \rho Ahg/A = \rho gh</math>) should be done step-by-step on the board, with students filling in each step in their notebooks. This derivation connects <math>P = F/A</math> (Lesson 2) directly to <math>P = \rho gh</math>.</p>                                                                                                                                                                            | <p>pressures in dams, water tanks, the Indian Ocean, and Lake Victoria. They also calculated total pressure at depth (<math>P_{\text{total}} = P_{\text{atmospheric}} + \rho gh</math>). Teacher note: The hydrostatic paradox (that container shape doesn't matter) is counterintuitive and deserves explicit discussion. Ask: 'Would the pressure at 5 m depth be different in a wide lake vs. a narrow pipe?' — the answer is NO, and students should justify this using the formula.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>all directions, and this pressure can do enormous work. DQB update: Students revisited all DQB questions and marked which are now fully resolved. Teacher note: This is the penultimate lesson — push students to identify any remaining gaps before the final synthesis lesson. Prompt model revision to ensure <math>P = \rho gh</math> is integrated alongside <math>P = F/A</math> and Pascal's Principle.</p>                                                                                                                                                                                                                                        |
| <p><b>Lesson 9</b><br/><b>Unit Synthesis: Solving the Jerrycan Mystery and Connecting Pressure to Kenya's World</b></p> | <p>Students returned to the original anchoring phenomenon — the crushed jerrycan — and presented their FINAL, fully-developed explanatory models. They also analysed a new, unfamiliar scenario: why does a deep-sea fish brought to the surface die or deform? And why do mountaineers on Mt. Kenya need supplemental oxygen? These transfer tasks tested whether students could apply their full conceptual toolkit to novel situations. Teacher note: Cold-call at least 6 students to present specific parts of the jerrycan explanation — assign roles (atmospheric pressure expert, particle model expert, <math>P = F/A</math> calculator, Pascal's Principle connector). Celebrate the intellectual journey from 'mystery' to 'full explanation.'</p> | <p>All forms of pressure — atmospheric, liquid, and transmitted fluid pressure — are unified by the same underlying physics: fluids (gases and liquids) exert force per unit area in all directions. The crushed jerrycan is explained by: (1) steam condensing → fewer gas particles inside → lower internal pressure (particle model + Boyle's Law); (2) external atmospheric pressure (~101.3 kPa) remained unchanged → net inward force; (3) <math>P = F/A</math> quantifies the crushing force; (4) Pascal's Principle shows pressure was transmitted uniformly, causing symmetric collapse. Students also articulated real-world applications: hydraulic brakes in matatus, water towers in Nairobi, dam design on the Tana River, blood pressure in the human body. Teacher note: Ensure every student can verbally explain the jerrycan phenomenon using at least THREE of the unit's key concepts. This is the core competency-based outcome.</p> | <p>Students completed their Summary Table reflection questions and submitted their final cumulative models. The DQB was reviewed as a class — all major questions posted at the start of the unit should now have evidence-based answers. Teacher note: Photograph the final DQB and each student's final model for assessment purposes. Distribute the Summary Table reflection prompts and allow 10 minutes of silent individual writing. Close the unit by asking: 'The jerrycan crushed itself on Day 1. Today you can explain EXACTLY why. What does that tell you about the power of physics?' — allow open responses to celebrate student growth.</p> |